

Questions on the manuscript and the talk

2 October 2026

Nine questions, each answered from an artifact in the repository rather than from recollection. Sections 1 and 2 concern one sentence of section 4 of the manuscript; sections 3 and 4 concern one sentence of the speaker notes; section 5 concerns the extremal solutions of [arXiv:2303.12471](#); section 6 whether our solver could construct them; section 7 the mass-curve secants of the result slide; and section 8 why Ψ is computed at all when the extremal mass comes from the asymptotic charges, and what comparing the two routes does and does not test; and section 9 how the extrapolation to $T = 0$ is done, and what happens to the comparison of section 8 when Ψ and μ get the same fit treatment. What was *not* checked is stated at the end of each answer and collected under Provenance.

The framing used throughout, and in `slides-new` and `notes-new`: the extremal mass $M_{\text{ext}}(J, \alpha)$ and its slope are read from the asymptotic charges; Ψ is the potential conjugate to α , and its $T \rightarrow 0$ limit is the slope that the entropy of nearby states predicts (section 8). Where an answer below was first written under the older framing — Ψ as the better way to obtain the slope — it has been revised.

1 What does 356 mean, and are the couplings equispaced?

The sentence at issue opens section 4, *Results*:

“The dataset contains 356 accepted solutions at 13 couplings from $\alpha = 0$ to $\alpha = 1/2$ in units $r_H = 1$.”

The 356 is a total over couplings, not a count per coupling

It is the number of converged black-hole solutions *summed over all thirteen couplings* — not 356 per coupling, and nothing to do with the number of α values. Each coupling carries one *walk*: a continuation in the spin at that fixed α (section 2 below). The number 356 is the total length of the thirteen walks.

The counts are bimodal, and the split is exactly the one the next sentence of section 4 describes. The six couplings walked over the *whole* spin range, $\alpha = 0, 0.02, 0.05, 0.1, 0.2, 0.5$, carry 40–45 states each, 262 in total. The seven that start at $q = 0.6$ and run only to the near-extremal end carry 10–17 each, 94 in total. So

$$356 = \underbrace{262}_{6 \text{ full walks}} + \underbrace{94}_{7 \text{ narrow walks}}.$$

“Accepted” means the state passed the solver’s acceptance gate: the stored rows carry the diagnostics it is judged on (`max_relative_tensor_residual`, `psi_window_spread`, `mass_tail_spread`, and so on). Refused attempts are not in the count.

α	states	walk	$TJ^{1/3}$ range
0	43	full	1.12×10^{-5} to 8.74×10^{-2}
0.005	10	narrow	2.44×10^{-3} to 5.10×10^{-2}
0.01	11	narrow	1.21×10^{-3} to 5.14×10^{-2}
0.02	40	full	4.61×10^{-3} to 8.15×10^{-2}
0.035	11	narrow	3.76×10^{-3} to 5.13×10^{-2}
0.05	45	full	4.49×10^{-4} to 7.42×10^{-2}
0.075	16	narrow	4.38×10^{-4} to 4.75×10^{-2}
0.1	45	full	3.84×10^{-4} to 6.48×10^{-2}
0.15	17	narrow	2.57×10^{-4} to 3.90×10^{-2}
0.2	45	full	1.44×10^{-4} to 5.21×10^{-2}
0.3	16	narrow	1.06×10^{-4} to 2.59×10^{-2}
0.4	13	narrow	5.06×10^{-5} to 1.97×10^{-2}
0.5	44	full	3.34×10^{-6} to 3.34×10^{-2}
total	356		

Table 1: States per coupling, read from `results/egb-extremality.json`. The two populations are the two kinds of walk described in section 2.

The couplings are not equispaced

α	0	.005	.01	.02	.035	.05	.075	.1	.15	.2	.3	.4	.5
gap		.005	.005	.01	.015	.015	.025	.025	.05	.05	.1	.1	.1

The spacing is graded, widening by a factor of about 20 from the small- α end to the large- α end. That matches what is being measured: the result is the departure from the first-order value

$$M_{\text{ext}} = \frac{3}{2}\pi^{1/3}J^{2/3} + \pi\alpha + \mathcal{O}(\alpha^2),$$

so the $\alpha \rightarrow 0$ end has to be resolved finely enough to see the slope leave π , while the large- α end only has to track a slowly varying curve.

The grid is also aligned with its two uses. The six full walks sit at 0, 0.02, 0.05, 0.1, 0.2, 0.5, and the seven narrow ones interleave between them. The narrow walks exist only to add resolution to the extremal-slope curve, which is why they skip the low-spin region entirely.

2 What is a “walk”?

A *walk* is one continuation sequence: a chain of solutions at *fixed* α , stepped in the spin, each solve seeded by the previous one. It is the unit the dataset is organised into — thirteen couplings, thirteen walks, 356 states. What follows is read from `src/rotating_bh/extremality.py`.

Where a walk starts

`walk_to_extremality(alpha_gb)` first builds a base solution with `predictor_ladder(start=.60, ...)`: it climbs in the *coupling* from the known $\alpha = 0$ Myers–Perry solution up to the target α in steps of `coupling_step=.005`, holding the spin at $\Omega_H = 0.6$. In units $r_H = 1$ one has $q = r_H\Omega_H = \Omega_H$, which is why the manuscript calls this $q = 0.6$.

Where a walk goes

From that base it runs `walk` twice, in both directions in the spin, with `spin_step=.01` in Ω_H :

- `direction=-1`, down to $\Omega_H = 0$, the static limit;
- `direction=+1`, up until the temperature falls below `floor=1e-6`, the near-extremal end.

The walk is the concatenation `descending + [anchor] + ascending`, sorted in Ω_H . Its length is that coupling's row in the table above. This is precisely the six-versus-seven split: the couplings given both directions are the 40–45-state full walks, while the seven narrow ones were given only the ascending branch from $q = 0.6$ — the region the $T \rightarrow 0$ extrapolation actually consumes.

Why it is a walk and not a scan

Every step is Newton-solved from the previous solution as its seed. These states cannot be solved independently, because near extremality there is no good initial guess except a neighbour. Three behaviours of the loop follow from that, each recorded in the source as the answer to a way an earlier version failed:

- **Halve, do not stop.** A Newton failure is treated as a seeding accident rather than a physical boundary: the step halves and retries. What ends a walk is the step collapsing below `minimum_step=1e-5`. The docstring puts it plainly — “a wall which looked physical was not”.
- **Roll back on a refused state.** A state that misses the acceptance gate is not kept as the seed. Otherwise every later solve inherits the bad profile and the walk dies far short of where it could reach, which the source records as what the first production run did at the two smallest couplings.
- **Raise the resolution before shortening the step.** Approaching extremality the independent tensor residual grows, and the cure is $N = 64 \rightarrow 96$, not a smaller step. The resolution that accepted each state is stored with it, which is the `resolution` field in the rows.

Why 356 is not simply steps times couplings

A walk steps far more often than it records. States are kept on a schedule: whenever the spin has moved by `spin_record=.02` or the temperature has changed by a factor of `temperature_record=1.3`. Measuring at every accepted step would spend the run in `diagnose`; measuring on a fixed spin grid would crowd the samples where nothing changes. So 356 counts *recorded*, *accepted* states, and the number of Newton solves behind them is substantially larger.

This also explains the temperature coverage in the table of section 1. The $1.3\times$ temperature trigger makes the samples bunch up automatically as T collapses towards extremality — which is where `extremal_state` needs them, since it fits only the `window=12` coldest states.

3 What is the “rotating solver”?

The sentence at issue is in the speaker notes for the cost slide:

“Reproducing all known solutions took about four hours; the first new object, the rotating solver, took five times more ...”

It is the code that finds rotating Einstein–Gauss–Bonnet black holes: `src/rotating_bh/egb_rotating_bvp.py`, built in milestone (*hito*) 4. The ten-minute notes say it at greater length — “the first genuinely new object, the rotating Gauss–Bonnet black hole” — which is what the five-minute line compresses.

What it is

A spectral collocation solver for the radial boundary-value problem of the rotating EGB system. From its docstring:

- Chebyshev–Lobatto nodes; damped Newton with a *complex-step* Jacobian and a line search — the same architecture as the vacuum solver `vacuum_bvp.py`, deliberately mirrored.
- Four compact amplitudes B, F, H, W on a compactified radius, $z = 1/r$ and $x = 1 - z$, so that $x = 0$ is the horizon $r_H = 1$ and $x = 1$ is $r \rightarrow \infty$.
- The horizon row is singular in the generic interior formula — the coefficient matrix of the underlying $[f', b'', h'', w'']$ system degenerates there — so it uses separately derived horizon relations, verified against the exact Myers–Perry closed form.
- At infinity, $B = 1$, $F = 1$, $H_x = 0$, $W_x = 0$.
- At $\alpha_{\text{GB}} = 0$ the interior equations reproduce the vacuum ones exactly.

Everything in the paper rests on it: the walk of section 2 calls into it at every step, and the 356 states are its accepted outputs.

Why it was the expensive block

Hito 4 cost 11.2 hours and 508 M tokens, 31% of the project’s time and 36% of its tokens — the largest single block. The “five times” of the notes is a token ratio, and it is a comparison across different units:

Block	Hours	Tokens
Hitos 0–3, reproducing the known literature	4.4	101,558,821
Hito 4, rotating EGB black hole	11.2	507,954,588

$508/101.6 = 5.0$. Note that the sentence quotes hours for the first block and tokens for the second.

The ledger documents why it was costly:

- **The direct tensor derivation was symbolically intractable.** Christoffel and Riemann for the non-diagonal metric did not terminate. The work moved to the reduced action, with equivalence to the full tensor verified *numerically* over generic profiles rather than symbolically.
- **Five discarded solver routes** before the one that worked: fitting a single horizon parameter; multi-parameter shooting with `least_squares` and `solve_ivp`; `solve_bvp` seeded from Myers–Perry with an exact analytic Jacobian, which went singular with parameters diverging to 10^4 – 10^{22} ; extracting the horizon conditions from the compactified form, where `sp.limit` did not terminate; and translating the raw relations into compact variables, which gave correct but unrecognisable expressions that failed to reproduce even the vacuum form. The route that worked redid the horizon and infinity expansions *directly* in the compact variables.
- **Three self-inflicted errors**, each costed: z and x treated as independent symbols; the numerator extracted before the series, producing a spurious relation; and f'' , w'' omitted from the physical jets, which left an order-one tensor residual even on exact Myers–Perry.

4 Is it a different solver from the one of arXiv:1010.0860, and what was it used to reproduce there?

Different in method and independently written. The second half of the question has the more interesting answer: the rotating solver was never successfully checked against their *bulk numbers at finite coupling*. The single comparison that would do that is the one that failed.

The two solvers

	Brihaye, Kleihaus, Kunz, Radu arXiv:1010.0860	This project egb_rotating_bvp.py
Method	COLSYS, a standard off-the-shelf BVP package (their ref. [41]: Ascher, Christiansen & Russell)	Chebyshev–Lobatto spectral collocation, written from scratch
Nonlinear solve	Newton–Raphson with adaptive mesh selection	Damped Newton, complex-step Jacobian, line search
Grid	10^2 – 10^3 mesh points, adaptive	$N = 64$, raised to 96 on a refused state
Quoted accuracy	relative $\sim 10^{-6}$	tolerance 10^{-11} , with per-state residual gates
Equations	their own reduction	independently rederived via the reduced action, equivalence checked numerically

What the two share is a strategy, not code. They also continued from the closed-form solutions, “increasing gradually α or Ω_H ”, which is structurally the walk of section 2 together with the climb in the coupling that seeds it.

What was actually reproduced against them

The eight external contrasts of `results/egb-rotating-external-contrasts.json`, with their sectors, which are the point:

Check	Sector	Result
reduced normalisation pinned	convention	pass
$\omega_H^2(j^2)$ reduced relation	Einstein-limit	pass
each-spin reading of J	convention	pass
critical temperature	Einstein-limit	pass
Wald entropy matches paper	Gauss–Bonnet	pass
$b'(r_H) = 2r_H/(r_H^2 + \alpha)$ at $\Omega_H = 0$	Gauss–Bonnet	pass, $\sim 10^{-14}$
b', f' decrease monotonically in Ω_H	Gauss–Bonnet	pass
published ergosurface radii, their §4.1.1	Gauss–Bonnet	fail

To these the manuscript adds their near-horizon eqs. (4.11) and (4.12), satisfied to 9.17×10^{-14} and 6.66×10^{-16} , which is what pins the coupling conversion $\alpha_{\text{ref.}} = 4\alpha$.

The structure matters more than the tally. The file’s own scope note says the contrasts were chosen among statements “that are closed forms or carry more digits than the section 4.1.1 radii”. Hence:

- the Einstein-limit agreements **do not test the Gauss–Bonnet terms at all**, and the file labels them apart for that reason;
- the two tightest Gauss–Bonnet agreements are against *closed-form* statements — the static horizon-derivative formula at $\Omega_H = 0$, which carries no rotation, and the near-horizon algebra, which is computed by `near_horizon.py` and **does not use the bulk rotating solver**;

- the only check that compares our rotating bulk solutions at finite α against values they obtained *numerically* is the ergosurface radius, which gives 1.102101 against their 1.104 at $\alpha = 1/2$ in our convention. The manuscript states that the discrepancy is unresolved and declines to offer it as evidence of an erratum.

So, stated strictly: the rotating solver reproduces Myers–Perry exactly at $\alpha = 0$, reproduces closed-form EGB statements, and reproduces the near-horizon sector — but its finite- α rotating *bulk* output has one published cross-check, and that one disagrees in the fourth digit.

The manuscript’s containment argument is that the ergosurface radius tests the reconstruction of the metric functions *away from the horizon*, whereas the result depends on the asymptotic charge extraction, the response solve and the zero-temperature extrapolation, none of which uses r_e . That is a reasonable argument and it is made in the discussion rather than buried; whether it is strong enough is a judgement a referee may well press on.

5 Why do the extremal solutions of arXiv:2303.12471 not give M_{ext} , and with it an independent check?

In principle they do, and the manuscript says so: scale invariance makes the extremal branch one curve in their reduced variables, and that curve fixes $M_{\text{ext}}(J, \alpha)$ through $\mu = \mu(0) j^{-2/3}$ and $y = 4x\mu/(3\pi)$ (`manuscript/main.tex`, introduction). In practice the paper does not publish the branch in a form that can be compared at the precision of our results. Four separate reasons.

1. The near-horizon solution carries no mass

The exact rotating squashed $\text{AdS}_2 \times S^3$ solution (their eqs. (3.18)–(3.19)) gives J , A_H and $S(\text{extremal})$, not M . The mass is an asymptotic charge and the near-horizon geometry does not fix it; this is also why arXiv:1010.0860 obtained only the extremal entropy.

2. The full extremal solutions give M , but no numbers are printed

The directly constructed extremal solutions do give it, $M = \frac{3\pi^2}{4}(a - c_t)$ with c_t read off the large- r tail (their eq. (3.15)). The paper prints no table of c_t , M or J ; the branch appears only as curves in their Fig. 1, “resulting from the extrapolation of around one thousand of data points into the continuum”.

3. The figure does not show the projection we would need

Every quantity in Fig. 1 is normalised by M , and the extremal curve appears only as two separate projections: a_H against $x = c_0\alpha/M$ in the main panel, and a_H against j in the inset (likewise for s and t_H). There is no j against x , which is what fixes $\mu(y)$. Recovering it means matching the two projections point by point through a_H . As far as can be seen at the resolution inspected, the inset’s extremal curve is flat and not monotonic near $j \approx 0.95$ –1, so that matching is ill-conditioned there. That is where our branch lives: $j_{\text{ext}} = [\mu(0)/\mu]^{3/2}$ with μ rising by 31% gives j from 1 down to about 0.67.

4. Part of our result is a slope, and slopes read off a figure are the weak point

The headline numbers $\pi \rightarrow 1.07 \rightarrow 1.44$ are values of $\partial M_{\text{ext}}/\partial\alpha|_J$. Differencing masses divides their error by the spacing in the coupling. With our masses, good to 10^{-3} – 10^{-6} , that works (section 7); with masses read off a figure to about one percent, the slope would be uncertain by order one. The paper also states no accuracy for the equal-spin extremal solutions: the only

error estimate it gives ($\sim 10^{-3}$) is in its section 4, for the single-plane solutions. Even a perfect reading would have no error bar to compare against.

What could still be done

- *A coarse check is feasible.* The effects are large — the slope drops by a factor of about three and first order overshoots by $2.3\times$ — so a curve good to the percent level could confirm the sign and the overshoot.
- *Getting the curve.* The PDF’s Fig. 1 contains no raster images, so it is drawn as vector curves; the arXiv source may give the vertices exactly, as was done for `arXiv:1010.0860` (`short-summary/summary.tex`, “Reproducing the published family”).
- *The clean option.* Ask the authors for their extremal data (c_t , c_w , a at each α). The manuscript’s discussion already names a direct construction of the extremal branch as what would remove the extrapolation in temperature.

6 Can our solver find extremal solutions, and would constructing them be an independent check?

Not as it stands; a direct construction would be a worthwhile extension, though not a fully independent one.

Why the present solver cannot reach $T = 0$

The ansatz of `src/rotating_bh/egb_rotating_bvp.py` is built around a non-degenerate horizon: $b = (1 - z^2)B$ has a simple zero at $r_H = 1$, the horizon row uses regularity relations derived for that case (`_egb_rotating_horizon_generated.horizon`), and the inputs are (r_H, ω_H, α) . An extremal horizon is a double zero; the expansion there changes form and these relations no longer apply. That is why the project approaches $T = 0$ from above — the coldest state has $TJ^{1/3} \approx 3 \times 10^{-6}$ — and why the result rests on a fit in temperature.

`near_horizon.py` does solve an extremal problem, but only the near-horizon $\text{AdS}_2 \times S^3$ algebra (v_1, v_2, v_3, k) , not the full solution out to infinity. It therefore gives S_{ext} , not M_{ext} (section 5, point 1).

What the extension would involve

1. *A solver for the degenerate horizon*, for instance in the gauge of `arXiv:2303.12471` (horizon at $r = 0$, $u = r^2 + a^2$): a series expansion at the horizon whose leading data come from `near_horizon.py`, and the same asymptotic charge extraction as now.
2. *Seeds* from the coldest states of the existing walks, which at $TJ^{1/3} \sim 10^{-6}$ should already be very close.
3. *The slope at $T = 0$ directly*: the same implicit-differentiation response solve applied to the extremal family at fixed J , giving $\partial M_{\text{ext}}/\partial \alpha|_J$ in one linear solve, with no limit in temperature and no first-law argument at $T = 0$.

What it would and would not check

- *It removes the weakest link.* The $T \rightarrow 0$ extrapolation is what the error bars describe, and they are the spread over 21 fits, which cannot reveal an error shared by all of them. Comparing $\mu(y)$ and the slope at the 13 couplings with a direct construction would test it properly.

- *It is not fully independent.* It would reuse the same derived field equations and the same mass extraction, so an error in either would carry over. Independence from those needs the authors’ data (section 5) or a second-order perturbative calculation; the manuscript’s discussion names both, with the direct construction, as the three ways to sharpen the result.

The main technical risk

In higher-derivative theories the corrections near an extremal horizon can scale as non-integer powers of the distance to the horizon, so the solution is not smooth there; [arXiv:2303.07358](#) (in [papers/](#)) concerns this for Kerr. If it happens here, the horizon series needs those powers and the Chebyshev convergence degrades. [arXiv:2303.12471](#) states that it uses a power series at $r = 0$, which suggests the equal-spin case is regular, but this was not checked.

7 Do the mass-curve secants suffer from the finite-difference problem?

Yes. The open squares of the result slide are secants $\Delta\mu/\Delta y$ between neighbouring couplings, and they have the usual finite-difference trade-off; in this dataset it matters at the smallest couplings. The difference from the textbook case is the source of the noise: not machine rounding, but the error of each extrapolated extremal mass.

The two errors

For neighbouring couplings y_1, y_2 , with $\mu = M_{\text{ext}}/J^{2/3}$:

- *Noise.* Each $\mu(y)$ is a $T \rightarrow 0$ extrapolation with an error $\delta\mu$, so the secant carries roughly $(\delta\mu_1 + \delta\mu_2)/\Delta y$, which grows as Δy shrinks.
- *Truncation.* A secant is the slope averaged over the interval; evaluated at the midpoint it misses about $\mu''' \Delta y^2/24$, which shrinks with Δy .

So there is an optimal spacing, and below it the noise term blows up.

What the data show

From `results/egb-extremality.json`; the noise column is $(\delta\mu_1 + \delta\mu_2)/\Delta y$ from the stored `mu_spread`, and for the first interval it reproduces the stored `mass_only_spread` (0.263).

interval in y	Δy	$\delta\mu$ at the ends	secant noise	secant
0–0.005	0.0048	$1.3 \times 10^{-7}, 1.3 \times 10^{-3}$	0.26	3.07
0.005–0.010	0.0051	$1.3 \times 10^{-3}, 7.7 \times 10^{-4}$	0.40	2.91
0.010–0.021	0.0109	$7.7 \times 10^{-4}, 1.6 \times 10^{-3}$	0.22	2.65
0.021–0.038	0.0177	$1.6 \times 10^{-3}, 8.4 \times 10^{-4}$	0.14	2.22
0.038–0.057	0.0185	$8.4 \times 10^{-4}, 5.7 \times 10^{-6}$	0.05	1.82
$y > 0.057$	0.03–0.10	10^{-5} to 10^{-6}	≤ 0.002	1.47...1.39

- *Small y .* The couplings are closely spaced and their extrapolated masses are uncertain at the 10^{-3} level, so the secants are uncertain by 10–40%: the blow-up. That the squares land near the curve there is supported by the sampled values, not by their error estimates.
- *Larger y .* Masses are good to 10^{-5} – 10^{-6} and the spacing is wider, so the noise is negligible; the main error is the interval average, which is why the squares sit slightly off the dots around the minimum.

A rough optimum

Balancing the two errors gives $\Delta y_{\text{opt}} \approx (48 \delta\mu/|\mu''|)^{1/3}$. Near $y \approx 0.03$ the slope samples give $|\mu''|$ of a few hundred; with $\delta\mu \approx 10^{-3}$ that puts Δy_{opt} near 0.05, about ten times the spacing used there, and with $\delta\mu \approx 10^{-5}$ near 0.01. At small coupling the secants are noise-limited, and halving the spacing would make them worse.

Does Ψ avoid it?

Only in part, and only at small coupling. Ψ differences nothing in α : it comes from one linear solve with the Newton Jacobian at fixed α , so there is no $\Delta\alpha$ to choose (backup slide A2). Its error is the extrapolation of Ψ itself to $T = 0$. That is also largest at small coupling — the stored `psi_uncertainty` is 0.09–0.16 at $y = 0.005$ –0.04, against 0.001–0.003 at large y , because those couplings were walked over a narrow spin range and their coldest states are not very cold. In the first interval it is still smaller than the secant noise: 0.11 at the upper end, or 0.06 as the mean of the two ends used in section 8, against 0.26. With the stored spreads the secants look the more precise above $y \approx 0.07$, but those spreads give μ only three of the fits that Ψ gets. With the same fit list for both (section 9) the secant noise grows to 0.06–2.2 and Ψ is the more precise slope at every coupling, so the manuscript’s argument for measuring the derivative rather than differencing masses holds.

8 If M_{ext} comes from the asymptotics, why compute Ψ ?

Alert: parts of the manuscript need to be rewritten. The motivation for Ψ in the introduction of `manuscript/main.tex` (lines 124–131) says that differencing masses is unreliable and that Ψ gives the slope “measured rather than differenced”. With the stored spreads the data seemed to contradict this for $y \gtrsim 0.07$; with Ψ and μ on the same fit envelope (section 9) they support it at every coupling, so the precision argument stands and only its wording needs care. Three passages need revising: the introduction paragraph; the sentence “Figure 4 shows the mass that this integral accumulates” (line 972), since the plotted μ is the direct extrapolation and not the integral of Ψ_{ext} ; and the route comparison around line 1005, which should present the agreement as a check of the thermodynamic quantities and of the first-law subtraction, stating also what it cannot test. The squares of figure 5 and the points of figure 4 also lack error bars, although the spreads are stored. `slides-new` and `notes-new` already use the revised framing.

Bottom line (corrected, section 9): with the same 20-fit envelope for both, Ψ gives the slope more precisely than the mass read at infinity at every coupling. *Per interval* ($\int \Psi dy$ against $\Delta\mu$, and equally Ψ against the secant $\Delta\mu/\Delta y$, since both divide by the same Δy): Ψ has the smaller error **in all 12 intervals, by a factor 8–22**. *Cumulatively* ($\mu(0) + \int_0^y \Psi dy'$ against $\mu(y)$ read at infinity): the integral has the smaller error **for $y \leq 0.038$** (by a factor 4–20) and is **worse from $y = 0.057$ on** (by a factor 1.5–11), because the linearly summed errors of Ψ accumulate.

Superseded: an earlier version of this box said Ψ was better only for $y < 0.038$. That compared Ψ ’s wide envelope with the much narrower three-fit spread stored for μ , and is withdrawn.

Errors are half-widths of the fit-sensitivity envelope of section 9, computed by `checks/route_envelopes.py`, combined linearly; the cumulative Ψ error is their trapezoid sum from $y = 0$.

For the extremal mass curve itself, Ψ is not needed. Each solution’s mass is read from the asymptotic falloff, $b = 1 + U/r^2 + \dots$, $M = -3\pi U/8$, $J = \pi W/4$

(src/rotating_bh/egb_rotating_observables.py); M and J are extrapolated to $T = 0$ at each coupling (src/rotating_bh/extremality.py), and $\mu = M_{\text{ext}}/J_{\text{ext}}^{2/3}$. No integral of Ψ enters $\mu(y)$, and differencing $\mu(y)$ already gives the slope.

What the stored routes show

From route_comparison of results/egb-extremality.json; the Ψ spread is the stored goon_penco_spread, the mean of the two endpoint sensitivities. **Caution:** the two spread columns are not built the same way (section 9): the secant column carries only the three degree checks of μ , and the Ψ column an older two-model envelope narrower than the manuscript's. On equal footing the secant spreads grow by factors of about 5 to 900 and Ψ is the more precise slope throughout.

midpoint y	secant $\Delta\mu/\Delta y$	secant spread	Ψ interval mean	Ψ spread
0.0024	3.0671	0.26	3.0671	0.062
0.0073	2.9081	0.40	2.9081	0.10
0.0153	2.6463	0.22	2.6470	0.12
0.0296	2.2240	0.14	2.2258	0.13
0.0477	1.8188	0.046	1.8186	0.049
0.0723	1.4732	4.8×10^{-4}	1.4722	5.5×10^{-3}
0.1028	1.2365	1.5×10^{-3}	1.2353	1.6×10^{-3}
0.1465	1.1036	8.3×10^{-4}	1.1014	1.6×10^{-3}
0.2021	1.0758	3.6×10^{-4}	1.0745	2.6×10^{-3}
0.2791	1.1447	1.3×10^{-4}	1.1431	2.5×10^{-3}
0.3759	1.2712	9.7×10^{-5}	1.2706	2.4×10^{-3}
0.4669	1.3902	6.4×10^{-5}	1.3908	1.7×10^{-3}

- The central values agree to 10^{-5} – 10^{-3} everywhere.
- *With the stored spreads* the mass route looks more precise for $y \gtrsim 0.07$ and Ψ only for $y \lesssim 0.03$. *This is an artefact of the unequal envelopes and is withdrawn:* on equal footing Ψ is more precise in every interval, by a factor 8–22 (section 9).
- Ψ_{ext} is not “measured” at $T = 0$ either: it is a $T \rightarrow 0$ intercept with the same extrapolation step the manuscript names as its dominant error, and that step is where its sensitivity comes from.

So Ψ is the more precise slope, but not because it avoids the extrapolation: both routes extrapolate, and Ψ simply extrapolates more stably than the mass and spin do. The other points of this section — the mass curve does not need Ψ , and what the route comparison does and does not test — are unaffected.

What Ψ gives that the mass curve cannot

1. *A test of the extended first law with the Wald entropy for rotating EGB.* The mass route never uses S ; the Ψ route does, through $T \partial S / \partial \alpha$ in the subtraction, together with the regularity assumption $T \partial S_{\text{ext}} / \partial \alpha \rightarrow 0$. Agreement of the two routes is therefore a numerical check of the first law, the Jacobson–Myers entropy and the Goon–Penco relation for this family. It is a partial check: what it can and cannot test is set out below.
2. *Ψ at finite temperature over the whole domain* (figure 2): a thermodynamic response coefficient that no mass curve contains, and new for rotating solutions.
3. *A slope at each coupling without choosing neighbours.* This matters only for the derivative at a specific y , for example to locate the minimum, which the manuscript explicitly does not attempt.

First law and Smarr: one test, not two

- The Smarr relation $2M = 3TS + 6\Omega_H J + 2\alpha\Psi$ (`main.tex`, eq. `smarr`) is the first law combined with the scaling symmetry, which is exact for these solutions. Checking Smarr checks the first law along the scaling direction; the subtraction of eq. `psi` checks it along the coupling direction at fixed (r_H, Ω_H) . These are the same law along two paths.
- Smarr also gives Ψ algebraically from a single solution, with no derivative: $\Psi_{\text{Smarr}} = (2M - 3TS - 6\Omega_H J)/2\alpha$ (eq. `psismarr`). The manuscript already compares it with the implicitly differentiated Ψ and finds agreement at every finite coupling. If Ψ alone were the goal, Smarr would supply it without the linear solve. The implicit route earns its place where Smarr is weak — at small α the numerator is a near-cancellation divided by a small α — and it does not rely on the scaling argument.
- The extended first law is a theorem (Kastor et al.; Neri). Agreement therefore tests the implementation, not the law: the Wald/Jacobson–Myers entropy, the charge normalisations, the horizon quantities, and the assumption $T \partial S_{\text{ext}}/\partial\alpha \rightarrow 0$ at extremality. This is validation of the numerics, not a physics result.

What the route comparison does and does not test

It is a genuine consistency check, but a partial one.

- *What differs, and is therefore tested.* The mass route uses only M and J read at infinity. The Ψ route also uses T , the Wald entropy with its explicit α term, Ω_H , the α -derivatives from the linear solve, and the assumption $T \partial S_{\text{ext}}/\partial\alpha \rightarrow 0$; and it uses them along the fixed- (r_H, Ω_H) direction, while the mass curve runs along the extremal branch. Agreement of $\int \Psi dy$ with $\Delta\mu$ says that the numerical $(M, T, S, \Omega_H, J, \Psi)$ satisfy the extended first law consistently along two different paths through the solutions. That is not automatic: a wrong entropy term, a wrong T , a bad derivative or a wrong relative normalisation of M and S would each show up as a mismatch.
- *What is shared, and is therefore not tested.* The same solutions on the same branch, from the same field equations and discretisation (covered instead by the residual and resolution checks); and the $T \rightarrow 0$ extrapolation. M , J and Ψ are extrapolated separately, with the same kind of fit through the same cold states, so if the cubic misses the true approach to $T = 0$ all intercepts are biased.
- *Not “alike”.* Such a bias need not shift Ψ_{ext} by exactly the derivative of the shift in μ , so it could still produce a mismatch. But the errors can be correlated, and agreement does not bound them; the manuscript says the same of a common offset in the masses (`main.tex`, line 1031).

So the comparison checks the thermodynamic quantities and the first-law subtraction. It is not an independent measurement of M_{ext} and not a test of the extrapolation; only a solution built directly at $T = 0$ (section 6) removes that caveat. A direct test of the correlation is available: redo the route comparison for each of the 21 fit variants and see whether the two routes shift together.

What to claim as the physics content of Ψ

- Ψ itself over the rotating domain at finite temperature (figure 2), which has not been reported before.

- A non-perturbative check of the Goon–Penco relation for this family: the extremal mass slope equals $\lim_{T \rightarrow 0} (-T \partial S / \partial \alpha|_{M,J})$. This ties the sign of the mass shift to the sign of the entropy shift of nearby non-extremal states, a link the mass curve alone cannot make.

Suggested reframing

Keep the precision argument, but state it with the equal-footing envelopes of section 9, and do not present the extremal mass shift as the only reason to compute Ψ . Present Ψ as the potential conjugate to α , a measurable thermodynamic quantity, with the entropy–extremality relation of Goon and Penco as its application. Present the agreement with the first law and with Smarr as validation of the numerics, and say what that validation cannot reach: the shared solutions and the $T \rightarrow 0$ extrapolation. A referee reading the sensitivity budget next to figure 5 could ask exactly this question.

Not checked: whether the spreads of neighbouring μ are correlated; if they are, the secant spreads shrink further and the conclusion strengthens. The spreads are fit-sensitivity envelopes, not standard errors. The Smarr comparison was read from the manuscript, not recomputed; the statements on the first law, on Smarr and on what the route comparison tests are reasoned from the method and the manuscript’s equations. The ranges in the bottom-line box were computed here from those stored envelopes; with a quadrature instead of a linear combination they would shift somewhat. The 21-variant route comparison was not run, so whether the extrapolation errors of the two routes are correlated is open.

9 How is the extrapolation to $T = 0$ done?

No solution in the dataset has $T = 0$. The solver’s horizon ansatz is non-degenerate (section 6), so each walk climbs in spin towards extremality and stops cold but not at zero: $\tau = T J^{1/3} \leq 4.61 \times 10^{-3}$ at every coupling and 3.34×10^{-6} at the coldest. Every extremal number in the manuscript is the $T = 0$ intercept of a fit through those cold states.

The production fit

At each coupling the accepted states are sorted by the unscaled temperature T ($r_H = 1$). Each observable — M , J , S and Ψ separately — is fitted with a cubic polynomial in T through the six coldest states, and the constant term is its $T = 0$ value (`src/rotating_bh/extremality.py`, `extrapolate` and `extremal_state`). The invariants $y = \alpha/J_{\text{ext}}^{2/3}$, $\mu = M_{\text{ext}}/J_{\text{ext}}^{2/3}$ and $\sigma = S_{\text{ext}}/J_{\text{ext}}$ are then formed from these intercepts. The same code also fits degree 1 through four states and degree 2 through five; the range of the three intercepts is the stored `spread` of each observable (the “original degree checks”).

The sensitivity envelope

For Ψ the manuscript then asks how far the intercept moves when the fit is changed (`manuscript/reanalysis.py`, `sensitivity`): two coordinates (T and τ), two windows (the six and up to the twelve coldest states) and five models — polynomials of degree 1, 2, 3, $c_0 + c_1 v + c_{1/2} \sqrt{v}$ and $c_0 + c_1 v + c_2 v^2 + c_{\log} v^2 \log v$, with v the coordinate divided by its largest value in the window. That is 20 fits, the “21” of the manuscript counting the production one. The half-width is the largest distance of any intercept from the production value, or the degree spread if larger. Two further diagnostics: the production cubic fits its own six points to 1.77×10^{-6} , and dropping any one of them moves the intercept by at most 1.09×10^{-5} .

What it cannot see

The leading near-extremal powers of T were never derived, so the cubic is a convenience, not a model of the approach to extremality. The $\sqrt{v}$ and $v^2 \log v$ terms probe two specific departures from analyticity, not the general case. The envelope is a sensitivity to a stated list of choices: it carries no confidence level, and a bias common to all 20 fits — a wrong near-extremal form, or an error in the underlying charges — leaves it unchanged. The sensitivity is largest where the walk’s lever arm is shortest: the couplings 0.005–0.035 were walked over a narrow spin range and their coldest states span only a factor 12–42 in τ .

Two inconsistencies in how the envelopes are stored

- Ψ . The `psi_uncertainty` stored in `results/egb-extremality.json` is an older envelope: the degree spread or the shift from only the $\sqrt{T}$ and $T^2 \log T$ fits on the twelve coldest states, whichever is larger (`experiments/egb_extremality.py`). `manuscript/make_figures.py` replaces it by the 20-fit envelope before drawing the extremal table and figure 5 of the manuscript, so the manuscript is self-consistent; but `route_comparison` in the JSON, and the table of section 8, use the older and narrower one. Above $y \approx 0.05$ it is smaller by a factor 2–25.
- μ , y , σ , M , J . These carry only the three degree checks (manuscript, table `budget`). They never received the 20-fit treatment.

Both routes on the same footing

`checks/route_envelopes.py` applies the same 20 fits to M and J , forms μ fit by fit, and compares. The μ envelope grows from 10^{-7} – 10^{-3} to 4×10^{-4} – 2×10^{-2} :

interval in y	Ψ envelope	secant envelope	ratio	cumulative ratio at upper end
0–0.005	0.062	1.33	0.047	0.05
0.005–0.010	0.100	2.25	0.045	0.15
0.010–0.021	0.124	2.17	0.057	0.12
0.021–0.038	0.146	2.24	0.065	0.22
0.038–0.057	0.075	1.34	0.056	1.9
0.057–0.088	0.020	0.25	0.081	1.5
0.088–0.118	0.022	0.30	0.071	1.6
0.118–0.175	0.017	0.15	0.12	2.3
0.175–0.229	0.012	0.14	0.089	2.4
0.229–0.329	0.0087	0.065	0.13	3.6
0.329–0.423	0.0066	0.074	0.089	2.5
0.423–0.511	0.0042	0.058	0.072	11

The ratio is Ψ envelope over secant envelope (below one: Ψ better). The cumulative ratio compares the trapezoid sum of the Ψ envelope from $y = 0$ with the μ envelope at the upper end.

This reverses the comparison of section 8. With the same fits for both, Ψ gives the slope 8–22 times more precisely than the mass secants in every interval. The mass curve itself, $\mu(y)$, is still better read directly than integrated from Ψ beyond $y \approx 0.05$. The earlier “mass route wins above $y \approx 0.07$ ” came from giving μ only three fits. The manuscript has the same blind spot: its $\delta\mu$ column, and the combined spread in its route comparison, use the three-fit spread.

Not checked: the 20-fit list is the manuscript’s choice and was applied to M and J unchanged; M and J were paired fit by fit, which keeps their correlation, but no other pairing was tried.

Whether the shifts of Ψ and of the μ secants are correlated across the 20 fits — the test proposed in section 8 — was not computed; the intercepts needed for it are now available from the same script.

Provenance

The per-coupling counts, the α grid and the temperature ranges were read directly from `results/egb-extremality.json` and reproduce the generated macros `\Nstates = 356`, `\Ncouplings = 13`, `\Nmapcouplings = 6`, `\Nnarrowcouplings = 7` and `\Nmaplist` of `manuscript/numbers.tex`. The description of the walk is read from `src/rotating_bh/extremality.py`.

The description of the rotating solver is read from the docstring of `src/rotating_bh/egb_rotating_bvp.py`; the costs, the five discarded routes and the three errors from `prompts/uso-de-tiempo-y-tokens.md`, sections 2 and 3. The method of `arXiv:1010.0860` is from its section 4.1.1 and its reference [41], in the copy under `papers/`. The contrast table is read from `results/egb-rotating-external-contrasts.json`; the ergosurface numbers are from the discussion section of `manuscript/main.tex`. Section 5 reads eq. (3.15), section 3.2, eqs. (3.18)–(3.20), Fig. 1 and the error statement of section 4.1 of `arXiv:2303.12471`, in the copy under `papers/`; the relations $\mu = \mu(0)j^{-2/3}$ and $y = 4x\mu/(3\pi)$ and the 31% growth are from `manuscript/main.tex`. That near-horizon data cannot fix an asymptotic charge is standard background, recalled rather than checked here. Section 6 reads the docstring and residual of `src/rotating_bh/egb_rotating_bvp.py`, the module list of `src/rotating_bh/`, the coldest temperature and the 21 fits from the speaker notes, and the closing paragraph of the discussion of `manuscript/main.tex`. The non-smoothness of extremal horizons in higher-derivative theories is recalled, not read from `arXiv:2303.07358`. Section 7 reads `mu`, `mu_spread`, `psi_uncertainty` and `y` of each extremal record, and `mass_only` and `mass_only_spread` of `route_comparison`, from `results/egb-extremality.json`. Section 8 reads `mass_only`, `mass_only_spread`, `goon_penco` and `goon_penco_spread` of `route_comparison` from the same file, the mass extraction from `src/rotating_bh/egb_rotating_observables.py` and `extremality.py`, the plotting code from `manuscript/make_figures.py`, and the quoted passages, the Smarr relation, eq. `psismarr` and its comparison from `manuscript/main.tex`. Section 9 reads `extrapolate` and `extremal_state` of `src/rotating_bh/extremality.py`, `sensitivity` of `manuscript/reanalysis.py`, the override in `manuscript/make_figures.py`, the construction of `psi_uncertainty` in `experiments/egb_extremality.py` and the numerics appendix of `manuscript/main.tex`; its numbers are the output of `checks/route_envelopes.py`, run on 2 October 2026.

Not checked: no walk was re-run and no solve was performed; the 356 stored rows were not individually re-tested against the acceptance thresholds, and the external contrasts were read from the stored artifact rather than recomputed. The ratio $508/101.6 = 5.0$ of section 3 is arithmetic on the ledger table; the notes do not show the division. For section 5, nothing was digitised and no vector extraction of Fig. 1 was attempted; the shape of the inset curve near $j = 1$ was judged by eye from a 110 dpi rendering of the page. Section 6 is a proposal: no extremal solve was attempted, the horizon expansion at a double zero was not derived, and whether the equal-spin extremal horizon is smooth was not checked. In section 7 the noise column and the optimal spacing are estimates made here, not results of the project; $|\mu'''|$ is read roughly from differences of the slope samples, and no secants were recomputed at other spacings.